

Joshua Alvarez

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Software engineer and product-focused builder experienced in full-stack applications, interactive interfaces, database-backed systems, and hardware-integrated software. Strong foundation in TypeScript, React, Java, SQL, debugging, and building reliable, intuitive products.

EDUCATION

Grand Canyon University

Bachelor of Science in Software Development · Honors College · GPA: 3.75

Phoenix, AZ

Sep. 2023 – Apr. 2026

EXPERIENCE

Undergraduate Teaching Assistant

Sep. 2024 – Apr. 2026

Grand Canyon University

Phoenix, AZ

- Supported students in developing and debugging applications using JavaScript, React, Python, Java, SQL, REST APIs, authentication, and relational databases
- Reviewed code for correctness, maintainability, edge cases, testing, and alignment with functional requirements
- Identified root causes across routing, application state, forms, database queries, authentication flows, and frontend/backend integration while communicating solutions clearly

Software Engineering Team Lead

Jan. 2025 – Apr. 2026

Coding With My Girls

Phoenix, AZ

- Led students through software projects covering programming fundamentals, web development, implementation planning, testing, and iterative problem-solving
- Translated project requirements into manageable technical tasks and mentored students through setup, logic, data-flow, and debugging challenges

TECHNICAL PROJECTS

Interval Ear-Training Platform | *Next.js, React, TypeScript, Tone.js, Web Audio API*

Jun. 2026 – Present

- Developing an interactive music-practice platform with interval, pitch, harmony, clef, keyboard, circle-of-fifths, and tuner experiences
- Built randomized question generation, real-time audio playback, scoring, animated feedback, difficulty-based behavior, and responsive piano-keyboard interactions
- Designed reusable game-state and audio-control systems to coordinate playback, user input, feedback timing, and cancellation across interactive exercises

Drift Music Platform | *Next.js, React, TypeScript, Node.js, PostgreSQL, OAuth*

Sep. 2025 – Dec. 2025

- Built and deployed a full-stack music platform with GitHub OAuth, role-based permissions, administrative review tools, and authenticated CRUD workflows
- Designed relational data models and implemented responsive interfaces, server-side operations, database queries, routing, validation, and error handling

Music-Synchronized Water Fountain | *JavaScript, Arduino, FFT, Kotlin, React Native*

Aug. 2025 – Apr. 2026

- Developed browser-based software that analyzed live audio frequencies and translated them into synchronized lighting, pump, and water effects
- Integrated a web interface with an Arduino Mega through serial communication and resolved timing, signal, audio-analysis, and device-response issues across the system

LEADERSHIP & ACTIVITIES

AWS Cloudathon | *AWS, Cloud Architecture, Solution Design, Cost Analysis*

Apr. 2025

- Collaborated in a 24-hour challenge to design an AWS-based technical solution for a client-defined business problem
- Helped develop the proposed architecture, implementation strategy, and cost estimate, then presented the solution to AWS representatives and client stakeholders

SKILLS

Languages: TypeScript, JavaScript, Java, Python, SQL, C#, C/C++, HTML/CSS

Frameworks & Technologies: React, Next.js, Node.js, Spring Boot, REST APIs, PostgreSQL, MySQL, OAuth, Web Audio API

Tools & Practices: Git, GitHub, AWS, Docker, Vercel, Render, Linux, Arduino, Testing, Debugging, Code Review, UI/UX Design